

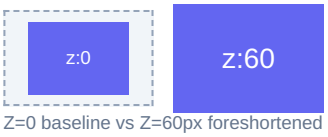
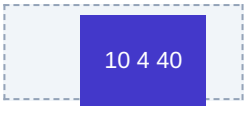


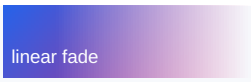
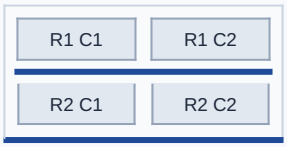
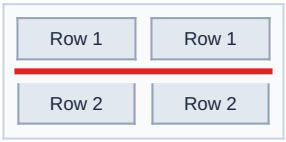
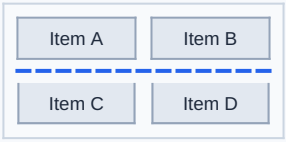
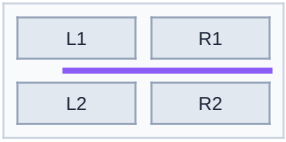
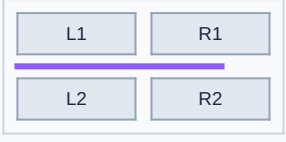
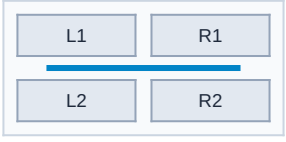
# Properties 390-420: 3D Transforms, Gradient Masks, and Row Rules

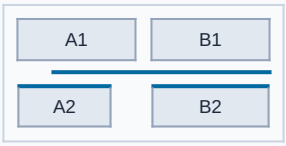
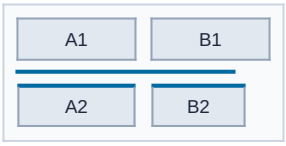
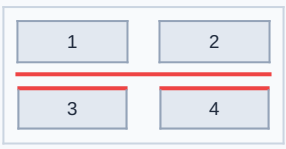
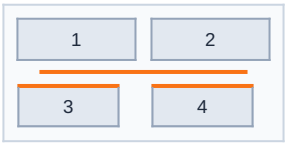
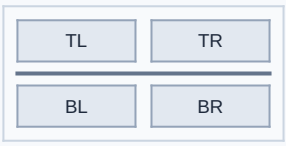
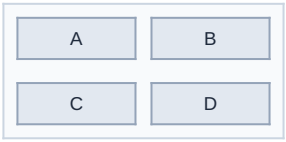
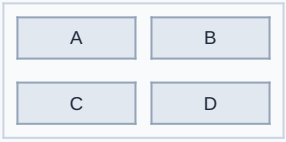
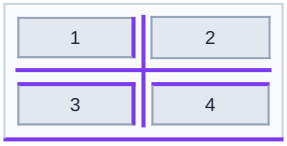
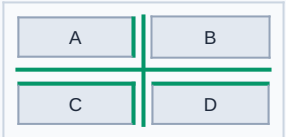
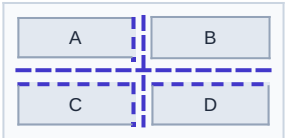
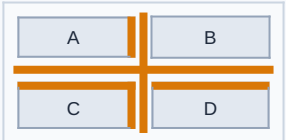
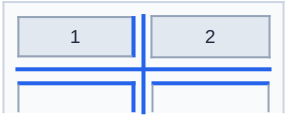
Properties added after baseline 389 · CSS Transforms, Masking, and Gap Rules · font-family "Liberation Sans"

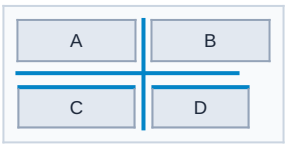
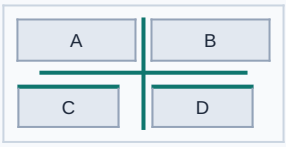
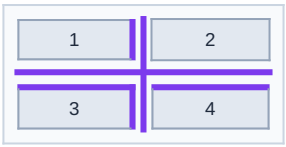
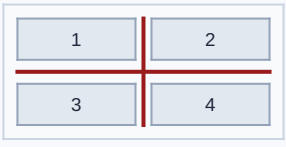
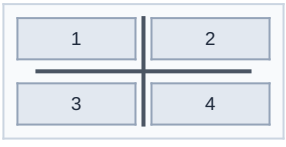
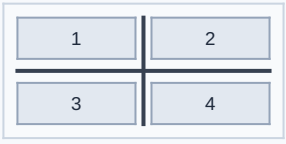
PROPS-390-TO-420

Audit of CSS properties and features implemented after baseline 389. Rows cover WebKit 3D prefix aliases, multi-axis 3D transform functions, gradient and image-set masks, and row/universal gap rules. Each row features an imperative specification and a live visual Effect demo.

| # | Property | What it should do | Effect (live) |
|---|---|---|---|
| 1 | `-webkit-backface-visibility`<br>legacy-vendor-alias · CSS Transforms · Chrome yes<br>Chrome supported | Maps to `backface-visibility`. Chrome culls the rotated back face on screen but paints it in print-to-PDF. |  <br>Chrome print shows both cards; screen culls the hidden back face |
| 2 | `-webkit-perspective`<br>legacy-vendor-alias · CSS Transforms · Chrome yes<br>Chrome supported | Maps to `perspective`. Establishes a 3D perspective projection distance for child elements rotated in 3D. |  <br>flat (orthographic) vs 180px perspective camera |
| 3 | `-webkit-perspective-origin`<br>legacy-vendor-alias · CSS Transforms · Chrome yes<br>Chrome supported | Maps to `perspective-origin`. Controls the center vanishing point for 3D perspective projection. |  <br>vanishing point left (10%) vs right (90%) |
| 4 | `-webkit-transform-style`<br>legacy-vendor-alias · CSS Transforms · Chrome yes<br>Chrome supported | Maps to `transform-style`. `preserve-3d` preserves shared 3D space across nested elements without 2D flattening. |  <br>flat intermediate vs tilted preserve-3d shared space |
| 5 | `translate3d()`<br>3D transform function · CSS Transforms · Chrome yes<br>Chrome supported | Translates an element in 3D along X, Y, and Z axes. Under perspective, positive Z brings the element closer to the viewer. |  <br>Z=0 baseline vs Z=60px foreshortened closer |
| 6 | `scale3d()`<br>3D transform function · CSS Transforms · Chrome yes<br>Chrome supported | Applies 3D scaling factors (sx, sy, sz) across width, height, and depth dimensions. |  |
| 7 | `translate (3-value)`<br>independent transform longhand · CSS Transforms · Chrome yes<br>Chrome supported | Accepts three values `x y z` to independently position an element in 3D without invoking `transform`. |  |
| 8 | `scale (3-value)`<br>independent transform longhand · CSS Transforms · Chrome yes<br>Chrome supported | Accepts three values `sx sy sz` to scale along width, height, and depth axes independently. |  |
| 9 | `rotate (axis vector)`<br>independent transform longhand · CSS Transforms · Chrome yes<br>Chrome supported | Accepts 4 tokens `x y z angle` (e.g. `rotate: 1 1 0 45deg`) to rotate around an arbitrary 3D vector. |  |

| | | | |
|---|---|---|---|
| 10 | `mask-image (linear-gradient)`<br>longhand · CSS Masking · Chrome yes<br>Chrome supported | Accepts `linear-gradient(...)` to smoothly feather or fade element painting into transparency. |  |
| 11 | `mask-image (repeating-linear-gradient)`<br>longhand · CSS Masking · Chrome yes<br>Chrome supported | Accepts `repeating-linear-gradient(...)` to mask an element with repeating stripes or venetian-blind patterns. |  |
| 12 | `mask-image (image-set)`<br>longhand · CSS Masking · Chrome yes<br>Chrome supported | Accepts `image-set(...)` candidate lists in mask layers, resolving the standard 1x candidate. |  |
| 13 | `mask (gradient shorthand)`<br>shorthand · CSS Masking · Chrome yes<br>Chrome supported | Parses gradient sources directly in the shorthand while resetting mask-border properties. |  |
| 14 | `row-rule`<br>shorthand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Shorthand setting width, style, and color for rule divider lines painted in grid or flex row gaps. |  |
| 15 | `row-rule-color`<br>longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Specifies the color of row rule divider lines painted between grid and flex rows. |  |
| 16 | `row-rule-style`<br>longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Specifies the stroke style of row rules ( `solid`, `dashed`, `dotted`, `double`). |  |
| 17 | `row-rule-width`<br>longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Specifies the thickness of rule lines painted inside row gaps. |  |
| 18 | `row-rule-visibility-items`<br>longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Controls whether row rules paint only between grid rows (`between`) or also at outer edges (`all`, `around`). |  <br>between: interior line; all: outer edges too |
| 19 | `row-rule-inset-start`<br>longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Insets the start (inline start / left) of the row rule divider line from the container margin. |  <br>line indented 24px from left |
| 20 | `row-rule-inset-end`<br>longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Insets the end (inline end / right) of the row rule divider line so it stops before the right boundary. |  <br>line stops 24px before right edge |
| 21 | `row-rule-inset-cap`<br>shorthand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Shorthand setting equal start and end insets for rule lines at outer cap edges. |  |

| | | | |
|---|---|---|---|
| | | | 16px cap inset on both ends |
| 22 | `row-rule-inset-cap-start`<br>longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Sets the inline-start cap inset specifically for row rule lines. |  |
| 23 | `row-rule-inset-cap-end`<br>longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Sets the inline-end cap inset specifically for row rule lines. |  |
| 24 | `row-rule-break`<br>longhand (partial) · CSS Gap Rules · Chrome experimental<br>Partial support | Configures break behavior (e.g. `intersection`) where row rules intersect crossing column gaps. |  <br>break across column gap |
| 25 | `row-rule-inset`<br>shorthand / longhand (partial) · CSS Gap Rules · Chrome experimental<br>Partial support | Sets start and end insets simultaneously for row rules. |  |
| 26 | `row-rule-inset-junction`<br>shorthand (partial) · CSS Gap Rules · Chrome experimental<br>Partial support | Sets insets around intersection junctions where row and column rules cross. |  |
| 27 | `row-rule-inset-junction-start`<br>longhand (partial) · CSS Gap Rules · Chrome experimental<br>Partial support | Sets start gap offset around intersection junctions along row rules. |  |
| 28 | `row-rule-inset-junction-end`<br>longhand (partial) · CSS Gap Rules · Chrome experimental<br>Partial support | Sets end gap offset around intersection junctions along row rules. |  |
| 29 | `rule`<br>shorthand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Universal shorthand setting width, style, and color across both column rules and row rules simultaneously. |  |
| 30 | `rule-color`<br>shorthand / longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Sets the color for all gap divider lines across both row and column directions. |  |
| 31 | `rule-style`<br>shorthand / longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Sets the stroke style (`dashed`, `dotted`, `solid`) across both row and column gap rules. |  |
| 32 | `rule-width`<br>shorthand / longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Sets the line thickness for all gap rules in the container. |  |
| 33 | `rule-visibility-items`<br>shorthand / longhand · CSS Gap Rules · | Controls visibility of rules between items or around all perimeter edges. |  |

| | | | |
|---|---|---|---|
| | Chrome experimental<br>Chrome experimental | |  |
| 34 | `rule-inset-start`<br>shorthand / longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Sets start insets across both row rules and column rules simultaneously. |  |
| 35 | `rule-inset-end`<br>shorthand / longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Sets end insets across both row rules and column rules simultaneously. |  |
| 36 | `rule-inset-cap`<br>shorthand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Sets symmetric cap insets for all gap rules in both row and column directions. |  |
| 37 | `rule-overlap`<br>longhand · CSS Gap Rules · Chrome experimental<br>Chrome experimental | Determines how row and column rules overlap and composite when crossing at grid junctions. |  |
| 38 | `rule-break`<br>shorthand (partial) · CSS Gap Rules · Chrome experimental<br>Partial support | Sets break modes for both row rules and column rules at gap intersections. |  |
| 39 | `rule-inset`<br>shorthand (partial) · CSS Gap Rules · Chrome experimental<br>Partial support | Sets universal insets across start and end of all rules. |  |
| 40 | `rule-inset-junction`<br>shorthand (partial) · CSS Gap Rules · Chrome experimental<br>Partial support | Sets junction insets across all rule intersections in both directions. |  |

Source: blinkless CSS catalog audit (post-389 additions) · Pattern: fixture-60/61/62/64 · Properties 390-420+ (40 demonstrated rows) · Needle: PROPS-390-TO-420 · font-family "Liberation Sans"